



清华大学

习题 1.28

$$\begin{aligned}
 F^{-1}[\hat{f}(\omega)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \hat{f}(\omega) e^{-i\omega x} d\omega = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\tau) e^{-i\omega \tau} d\tau \right) e^{-i\omega x} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\xi) e^{-i\omega(\xi-x)} d\xi d\omega \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\xi) \left(\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-i\omega(\xi-x)} d\omega \right) d\xi \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\xi) \cdot \sqrt{2\pi} \cdot \delta(x-\xi) d\xi = f(x)
 \end{aligned}$$

习题 1.29

$$\begin{aligned}
 (1) \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\frac{e^{i\lambda x} - e^{-i\lambda x}}{2i} \right) e^{-i\omega x} dx = \frac{\lambda}{8\sqrt{2\pi}} \int_{-\infty}^{+\infty} (e^{i(3\lambda-x)} - 3e^{i\lambda x} + 3e^{i(-\lambda-x)} - e^{i(-3\lambda-x)}) e^{-i\omega x} dx \\
 &= \frac{\lambda}{8\sqrt{2\pi}} \cdot 2\pi (\delta(3-\omega) - 3\delta(1-\omega) + 3\delta(-1-\omega) - \delta(-3-\omega)) \\
 &= \frac{\lambda\sqrt{2\pi}}{8} (\delta(\omega-3) - 3\delta(\omega-1) + 3\delta(\omega+1) - \delta(\omega+3))
 \end{aligned}$$

$$(2) \hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} (1 + 2\delta(x)) F[1] - 2F[\delta(x)] + 3F[\delta'(x)] = \sqrt{2\pi} \delta(\omega) - \frac{2}{\sqrt{2\pi}} + \frac{3}{\sqrt{2\pi}} i\omega$$

$$\begin{aligned}
 (3) \hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x \cos(2x) H(x-2) e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} \int_2^{+\infty} x \cos(2x) e^{-i\omega x} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_2^{+\infty} x \left(\frac{e^{-i(\omega-2)x} + e^{-i(\omega+2)x}}{2} \right) dx
 \end{aligned}$$

$$\begin{aligned}
 F[H(x)] &= \sqrt{\frac{\pi}{2}} \delta(\omega) + \frac{1}{i\omega\sqrt{2\pi}} \\
 F[xH(x)] &= i \frac{d}{d\omega} \left(\sqrt{\frac{\pi}{2}} \delta(\omega) + \frac{1}{i\omega\sqrt{2\pi}} \right) = i\sqrt{\frac{\pi}{2}} \delta'(\omega) - \frac{1}{\sqrt{2\pi}\omega^2} \\
 &= \sqrt{\frac{\pi}{2}} \delta'(\omega) - \frac{1}{\sqrt{2\pi}\omega^2}
 \end{aligned}$$

$$\begin{aligned}
 F[xH(x-2)] &= F[(x-2)H(x-2)] + 2F[H(x-2)] \\
 &= e^{-i\cdot 2\omega} \left(i\sqrt{\frac{\pi}{2}} \delta'(\omega) - \frac{1}{\sqrt{2\pi}\omega^2} \right) + 2e^{-i\cdot 2\omega} \left(\sqrt{\frac{\pi}{2}} \delta(\omega) + \frac{1}{i\omega\sqrt{2\pi}} \right) = e^{-i\cdot 2\omega} \left(2\sqrt{\frac{\pi}{2}} \delta(\omega) - \frac{1}{\sqrt{2\pi}\omega^2} \right)
 \end{aligned}$$

$$\begin{aligned}\hat{f}(\omega) &= \mathcal{F}[\lambda \cos(2x) H(x-2)] = \frac{1}{2} \left(\mathcal{F}[\lambda H(x-2) e^{i \cdot 2x}] + \mathcal{F}[\lambda H(x-2) e^{i(-2x)}] \right) \\ &= \frac{1}{2} \left\{ e^{-i \cdot 2(\omega-2)} \left[\lambda \sqrt{\frac{\pi}{2}} \delta'(\omega-2) - \frac{1}{\sqrt{2\lambda(\omega-2)^2}} \right] + 2 e^{-i \cdot 2(\omega-2)} \left[\sqrt{\frac{\pi}{2}} \delta(\omega-2) + \frac{1}{i(\omega-2)\sqrt{2\pi}} \right] \right. \\ &\quad \left. + e^{-i \cdot 2(\omega+2)} \left[\lambda \sqrt{\frac{\pi}{2}} \delta'(\omega+2) - \frac{1}{\sqrt{2\lambda(\omega+2)^2}} \right] + 2 e^{-i \cdot 2(\omega+2)} \left[\sqrt{\frac{\pi}{2}} \delta(\omega+2) + \frac{1}{i(\omega+2)\sqrt{2\pi}} \right] \right\}\end{aligned}$$

習題 1.30

$$\begin{aligned}(1) \mathcal{L}[f(t-\tau)] &= \int_0^{+\infty} f(t-\tau) e^{-pt} dt = \int_{-\tau}^{+\infty} f(t) e^{-p(t+\tau)} dt = e^{-p\tau} \int_{-\tau}^{+\infty} f(t) e^{-pt} dt \\ &= e^{-p\tau} \int_0^{+\infty} f(t) e^{-pt} dt = e^{-p\tau} \bar{f}(p)\end{aligned}$$

$$(2) \mathcal{L}[f(\frac{t}{k})] = \int_0^{+\infty} f(\frac{t}{k}) e^{-pt} dt =$$

$$\text{當 } k > 0 \text{ 時, } = \int_0^{+\infty} f(\tau) e^{-p(\frac{\tau}{k})} d(\frac{\tau}{k}) = \frac{1}{k} \int_0^{+\infty} f(\tau) e^{-\frac{p}{k}\tau} d\tau = \frac{1}{k} \bar{f}(\frac{p}{k})$$

~~當 $k < 0$ 時~~

$$\begin{aligned}(3) \bar{f}^{(n)}(p) &= \frac{d^n}{dp^n} \int_0^{+\infty} f(t) e^{-pt} dt = \int_0^{+\infty} f(t) \left(\frac{d^n}{dp^n} e^{-pt} \right) dt = \int_0^{+\infty} f(t) \cdot (-t)^n e^{-pt} dt \\ &= \int_0^{+\infty} (-t)^n f(t) e^{-pt} dt = \mathcal{L}[(-t)^n f(t)]\end{aligned}$$

$$\begin{aligned}(4) \mathcal{L}\left[\int_0^t f(\tau) d\tau\right] &= \int_0^{+\infty} \left(\int_0^t f(\tau) d\tau\right) e^{-pt} dt = -\frac{1}{p} \left[\left(\int_0^t f(\tau) d\tau\right) e^{-pt} \right]_0^{+\infty} - \int_0^{+\infty} f(t) e^{-pt} dt \\ &= \frac{1}{p} \bar{f}(p)\end{aligned}$$

$$(5) \mathcal{L}\left[\frac{f(t)}{t}\right] = \int_0^{+\infty} \frac{f(t)}{t} e^{-pt} dt = \int_0^{+\infty} \frac{f(\sqrt{t})}{\sqrt{t}} e^{-p\sqrt{t}} d\sqrt{t}$$

$$\begin{aligned}\int_p^{\infty} \bar{f}(s) ds &= \int_p^{\infty} \left(\int_0^{\infty} f(t) e^{-st} dt \right) ds = \int_0^{\infty} \int_p^{\infty} f(t) e^{-st} ds dt = \int_0^{\infty} \frac{f(t)}{-t} \left(e^{-st} \right)_p^{\infty} dt \\ &= \int_0^{\infty} \frac{f(t)}{-t} (-e^{-pt}) dt = \int_0^{\infty} \frac{f(t)}{t} e^{-pt} dt = \mathcal{L}\left[\frac{f(t)}{t}\right]\end{aligned}$$



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习题 1.31

$$1) f(t) = \sinh(\alpha t) = \frac{e^{\alpha t} - e^{-\alpha t}}{2}, \quad \mathcal{L}[f(t)] = \frac{1}{2} \left(\frac{1}{p-\alpha} - \frac{1}{p+\alpha} \right) = \frac{\alpha}{p^2 - \alpha^2}$$

$$2) \mathcal{L}[f(t)] = \mathcal{L}[t^2] + 3\mathcal{L}[t] + 2\mathcal{L}[1]$$

$$= \frac{2!}{p^3} + 3 \cdot \frac{1!}{p^2} + 2 \cdot \frac{0!}{p^1} = \frac{2}{p^3} + \frac{3}{p^2} + \frac{2}{p}$$

$$3) \mathcal{L}[f(t)] = \frac{n!}{(p-\alpha)^{n+1}}, \quad 4) f(t) = t^2 \cdot \frac{e^{(-1+3i)t} - e^{(-1-3i)t}}{2i}, \quad \mathcal{L}[f(t)] = \frac{1}{2i} \left(\frac{2}{(p+1-3i)^3} - \frac{2}{(p+1+3i)^3} \right)$$

$$= \frac{2}{i} \cdot \frac{9i(p+1)^2 + 27(-i)}{[(p+1)^2 + 9]^3}$$

$$= \frac{18(p+1)^2 - 54}{[(p+1)^2 + 9]^3}$$

$$5) \mathcal{L}[f(t)] = \frac{1}{p} \mathcal{L}[te^{2t} \sin 3t] = \frac{1}{p} \cdot \frac{1}{2i} \cdot \left(\frac{1}{(p+2-3i)^2} - \frac{1}{(p+2+3i)^2} \right) = \frac{1}{2pi} \cdot \frac{4 \cdot 3i(p+2)}{[(p+2)^2 + 9]^2}$$

$$= \frac{6(p+2)}{p[(p+2)^2 + 9]^2}$$

$$6) \mathcal{L}[f(t)] = 6 \int_0^{\infty} e^{-pt} dt + 2 \int_0^2 e^{-pt} dt = \frac{6}{p} + \frac{2}{p} (1 - e^{-2p}) = \frac{8}{p} - \frac{2e^{-2p}}{p} = \frac{8 - 2e^{-2p}}{p}$$

习题 1.32

$$1) \mathcal{L}[1 - \cos t] = \frac{1}{p} - \frac{1}{2} \left(\frac{1}{p-i} + \frac{1}{p+i} \right) = \frac{1}{p} - \frac{p}{p^2+1}, \quad \mathcal{L}[(1 - \cos t)e^{-t}] = \frac{1}{p+1} - \frac{p+1}{(p+1)^2+1} \quad \begin{matrix} (p+1 > 0) \\ (p > 0) \end{matrix}$$

$$\mathcal{L}\left[\frac{1 - \cos t}{t}\right] = \int_p^{\infty} \left(\frac{1}{s} - \frac{s}{s^2+1} \right) ds$$

$$\mathcal{L}\left[\frac{1 - \cos t}{t} e^{-t}\right] = \int_p^{\infty} \left(\frac{1}{s+1} - \frac{s+1}{(s+1)^2+1} \right) ds$$

$$\Rightarrow \int_0^{\infty} \frac{1 - \cos t}{t} e^{-t} \cdot e^{0 \cdot t} dt = \int_0^{\infty} \left(\frac{1}{s+1} - \frac{s+1}{(s+1)^2+1} \right) ds = \left[\ln(s+1) - \frac{1}{2} \ln[(s+1)^2+1] \right]_0^{\infty}$$

$$= \frac{1}{2} \left[\ln \frac{(s+1)^2}{(s+1)^2+1} \right]_0^{\infty} = \frac{1}{2} \left(0 - \ln \frac{1}{2} \right) = \frac{\ln 2}{2}$$

$$(2) \int_0^{\infty} t^3 \frac{\sin t}{e^{-t}} dt = \int_0^{\infty} \frac{t^3}{2i} (e^{-(1-i)t} - e^{-(1+i)t}) dt = \frac{1}{2i} \cdot \left(\frac{3!}{(1-i)^4} - \frac{3!}{(1+i)^4} \right)$$

$$(2) \int_0^{\infty} t^3 e^{-t} \sin t dt = \operatorname{Im} \left[\int_0^{\infty} t^3 e^{-(1-i)t} dt \right] = \operatorname{Im} \left[\frac{3!}{(1-i)^4} \right] = \operatorname{Im} \left[\frac{3!}{4(1-i)} \right] = 0$$

$$(3) \int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \int_0^{\infty} \frac{\sin 2t}{t} dt =$$

$$\sim \left[\frac{\sin 2t}{t} \right] = \int_p^{\infty} \frac{2}{s^2+2^2} ds, \quad \text{for } p > 0, \quad \text{for } p < 0$$

$$= \frac{1}{2} \int_p^{\infty} \frac{1}{\left(\frac{s}{2}\right)^2 + 1} d\left(\frac{s}{2}\right) = \arctan\left(\frac{s}{2}\right) \Big|_p^{\infty} = \frac{\pi}{2} - \arctan\left(\frac{p}{2}\right),$$

$$\text{As } p \rightarrow 0, \quad \int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \arctan \frac{\infty}{2} - \arctan \frac{0}{2} = \frac{\pi}{2}$$

$$\lim_{p \rightarrow 0} \int_0^{\infty} \frac{\sin 2t}{t} e^{-pt} dt = \frac{\pi}{2}$$